Beginner Course on Microbial Genomics: Applications & Research (2026)

# Day 0

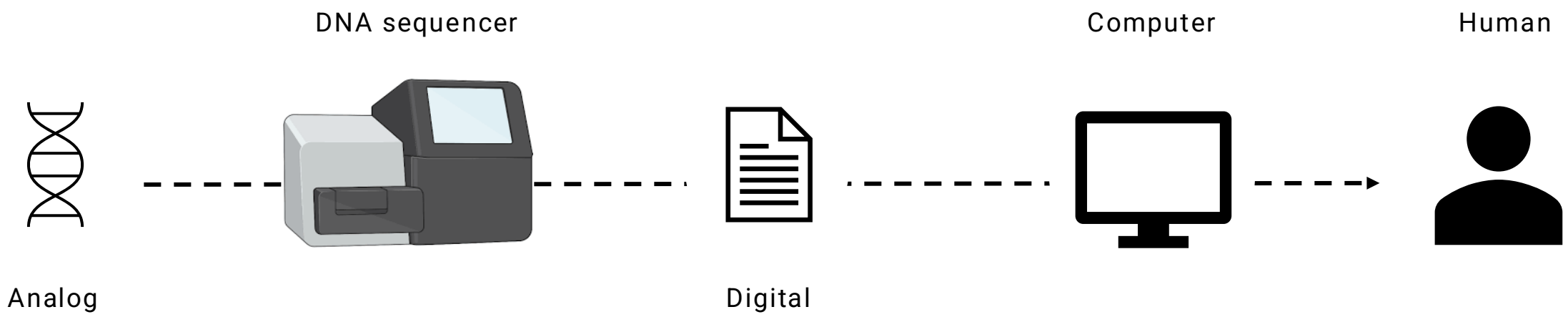
# Pre-workshop – Pack your backpack

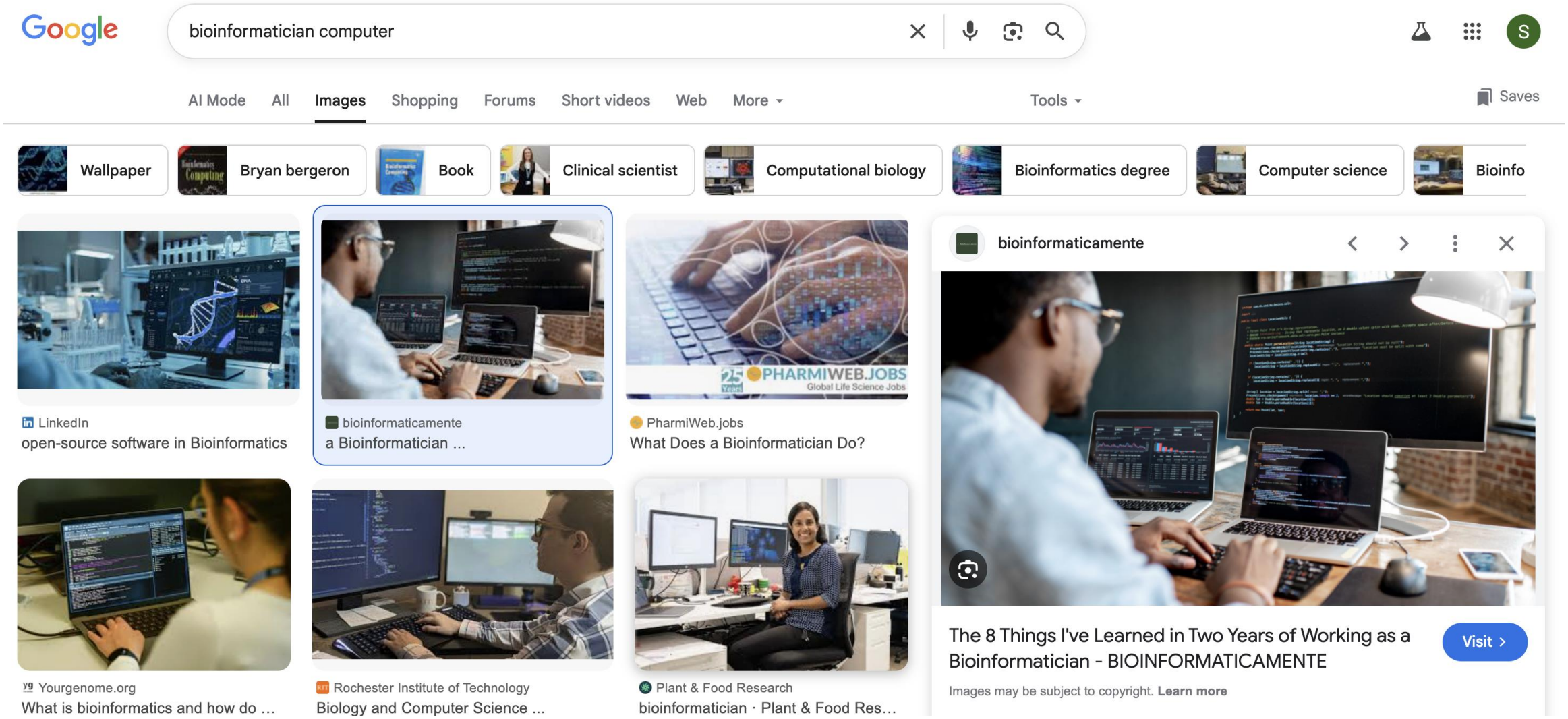
## INFORMATION

### Lecturers

Mai Thu Si Nguyen; Ton Ngoc Minh Quan

# Why are we here today?

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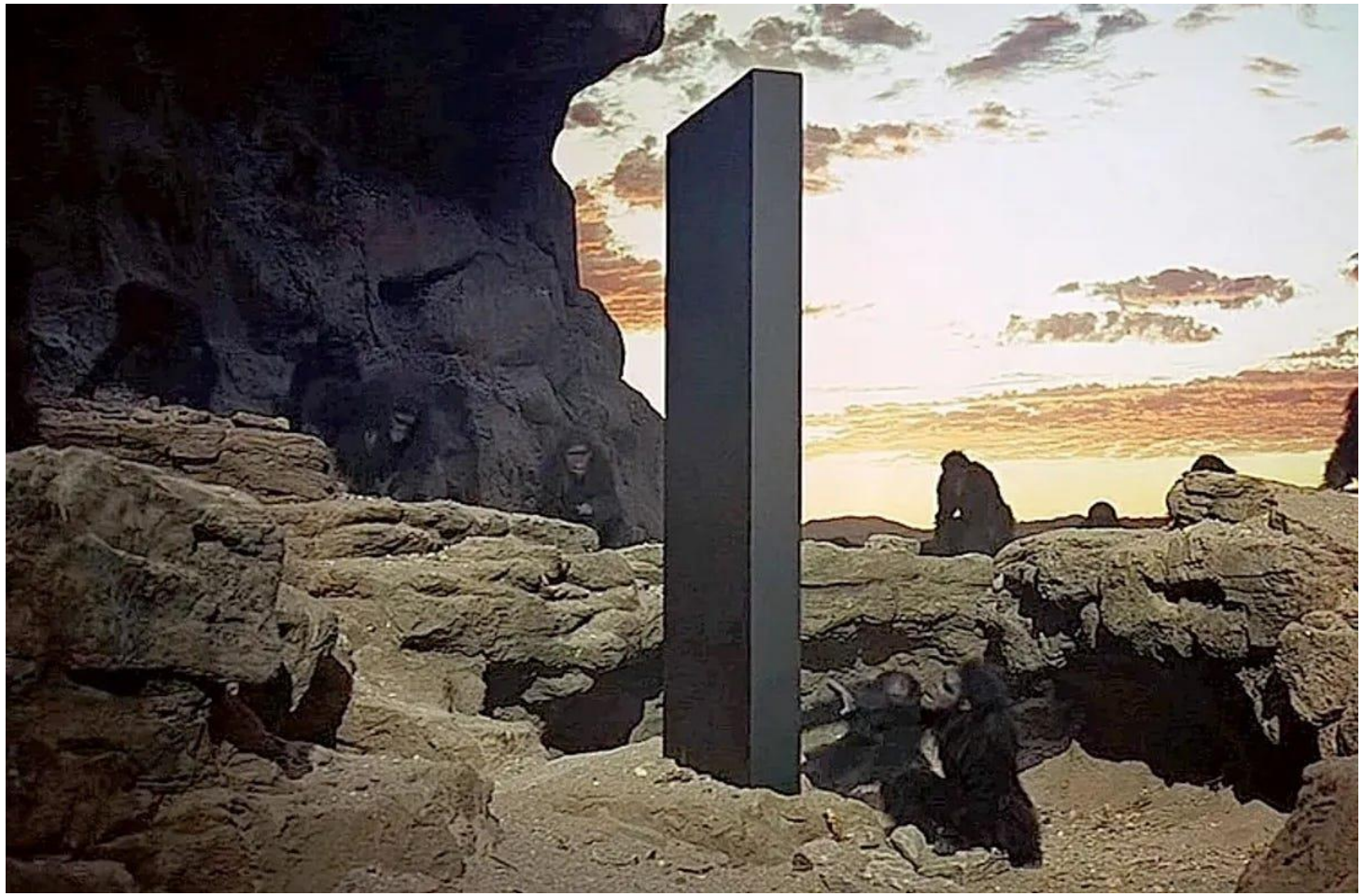
# Computer



***2001: A Space Odyssey.*** The monolith.

# Computer

***BBC Wildlife***

# Connect to the computer

# Setup your workspace

Kleb-CASAVA MicroGen 2026

Practical Guides | Lecture Slides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- Day 0 - Pre-Workshop
  - 0.1 Setup your workspace
    - Overview
    - Windows 10/11 setup
    - macOS setup
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    - Advanced R in VS Code
  - 0.2 Commands & Environments
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## 0.1 Setup your workspace


AUTHOR
Minh-Quan Ton-Ngoc



PUBLISHED
May 11, 2026



On this page
1 Purpose
2 Quick checklist
3 Which instructions should I follow?


## 1 Purpose

This guide helps you prepare a Windows or macOS laptop for the MicroGen 2026 workshop. It covers the tools used for coding, data analysis, and connecting to the workshop cloud server.

Most heavy bioinformatics workflows will run on the **workshop server**, not on your laptop. Your laptop mainly needs:

- a code editor,
- R and Python for small local exercises,
- SSH tools for connecting to the server,
- and optional graphical tools for users who prefer them.

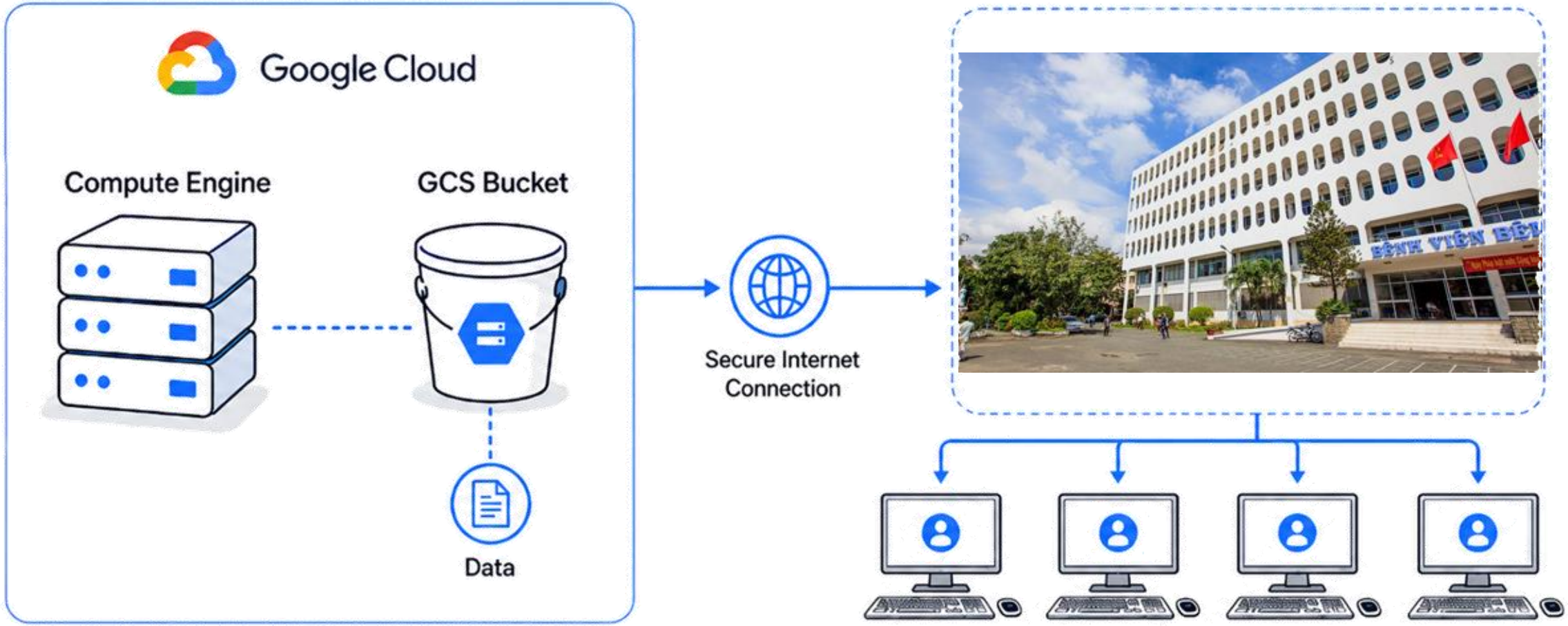
**Required and optional tools**

**Required for most participants:** Python 3.13, R 4.5.0, Visual Studio Code, the VS Code Remote - SSH extension, and the listed Python/R packages.

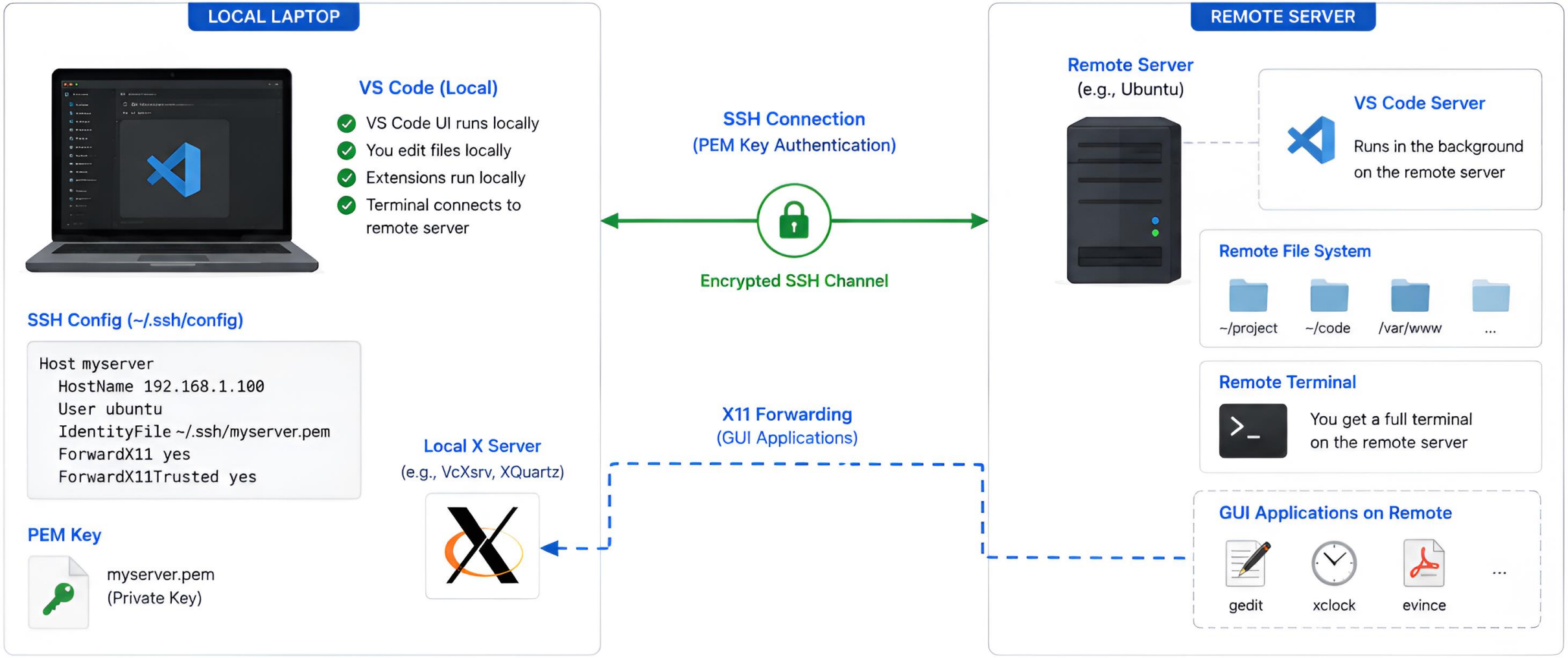
**Optional:** RStudio Desktop, Java, Artemis, BandageNG, SeaView, and extra X11/VNC tools. Instructors may ask you to use these for specific graphical activities.

**https://molepi-oucru.github.io/MicroGen-2026/day-0/instructions/setup-environment.html**

# Which computer?



# How to connect?



# How to connect?

You will need the following information to access the server through SSH:

- Username
- IP address
- PEM file: the key used to access the server through SSH.

**These details sent to you by email.**

Please note down this information and download the PEM file for later use.

For instructions on how to use the PEM file, please visit: **https://molepi-oucru.github.io/MicroGen-2026/day-0/instructions/setup-server-access.html**

# How to connect? - Troubleshooting

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## 0.1.7 Troubleshooting and FAQs


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On this page
1 Troubleshooting
2 FAQs
3 Further reading before the workshop

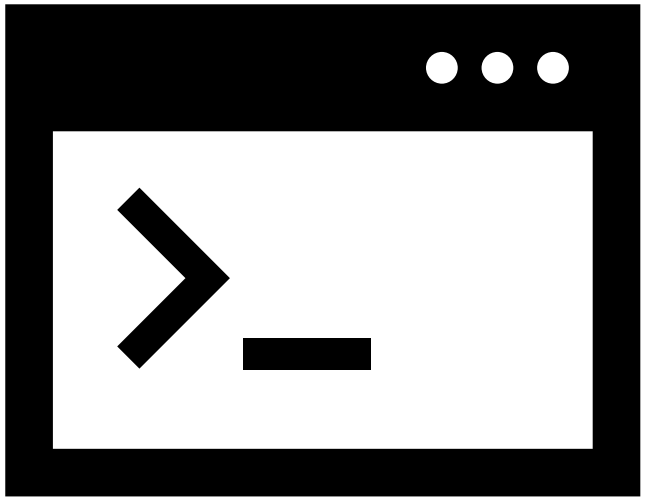

## 1 Troubleshooting

- `python` is not recognized on Windows
- `pip` is not recognized
- R package installation fails
- `phyloseq` installation fails
- Windows says Rtools is required
- macOS asks for command line tools
- macOS cannot verify that this app is free from malware
- VS Code cannot connect to the server

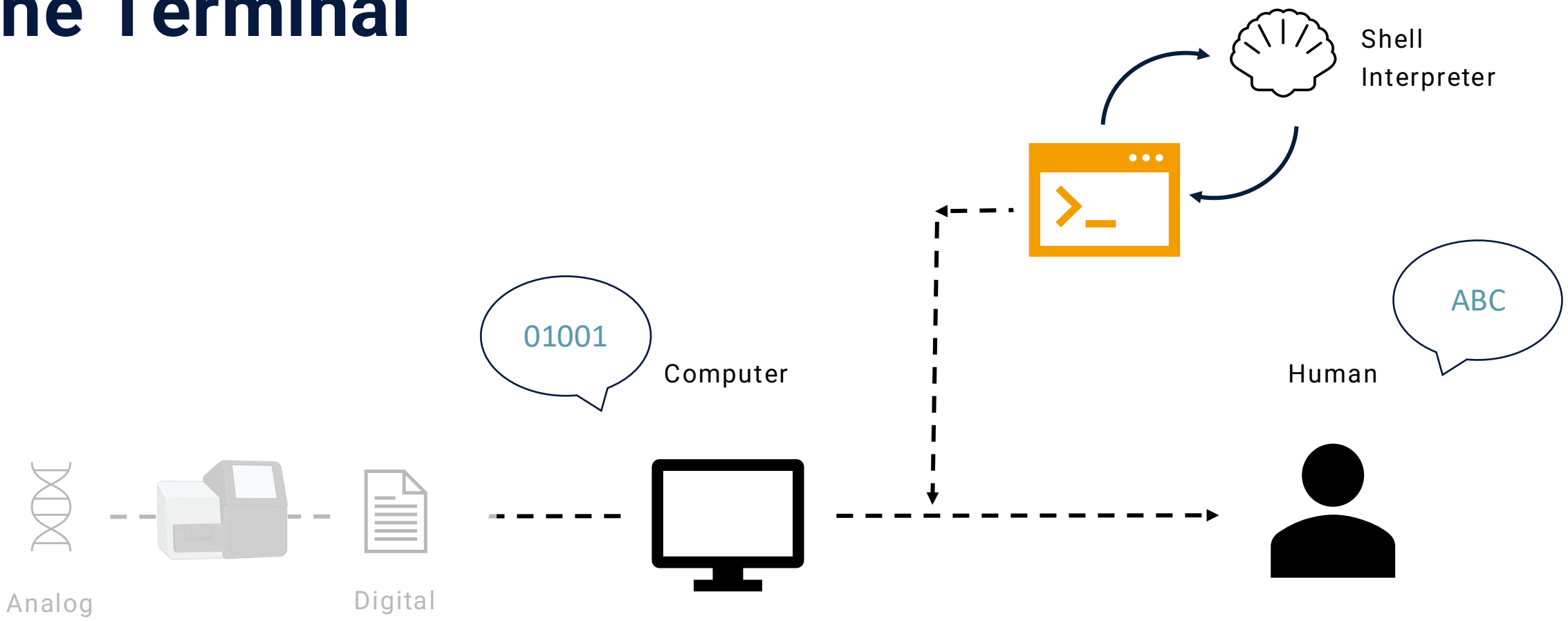
# Notes

- Each participant will receive an individual PEM file.
  - **Please DO NOT share this file with anyone outside the course.**
- If you would like **additional practice time after the class**: Please contact Quân and briefly explain your reason.
- If you need to continue using the server **after the course**:
  - *Access may be supported for up to 3–4 days after the course ends.*
  - Please contact Quân or Hao for assistance.
- If unauthorized sharing of server access is detected: **Organizing team reserves the right to suspend accounts and revoke server access without prior notice.**
- **DO NOT misuse the server or perform actions outside the instructor's guidance.** Exceptions are only allowed if explicitly approved by instructor(s) or teaching assistant(s).

# Data Analysis on Linux



# The Terminal



# The Terminal

User name

Computer name

Current **working directory**

# Your first command

*“ Know thyself”*

Problems Output Debug Console Terminal Ports 11 bash

```
(base) ubuntu@nguyenmts260124:/data/MicroGen-2026$ hostnamectl
   Static hostname: nguyenmts260124
         Icon name: computer-vm
           Chassis: vm
        Machine ID: 33c0ce0270234dc39e94b30638fc5b09
           Boot ID: 8ef34c30eb504cad849a004f2bbad2e9
    Virtualization: kvm
  Operating System: Ubuntu 20.04.6 LTS
            Kernel: Linux 4.15.0-213-generic
      Architecture: x86-64
(base) ubuntu@nguyenmts260124:/data/MicroGen-2026$
```

# Syntax

A command usually follows this structure: `command [options] [arguments]` input

flag

***Examples***:

Practice

```
# show Linux OS name, version, and hardware architecture
hostnamectl

# list CPU information
lscpu
```

Practice

```
echo "Hello World"
```

- `echo` is the command (print text).
- `"Hello World"` is the argument (the text to print).

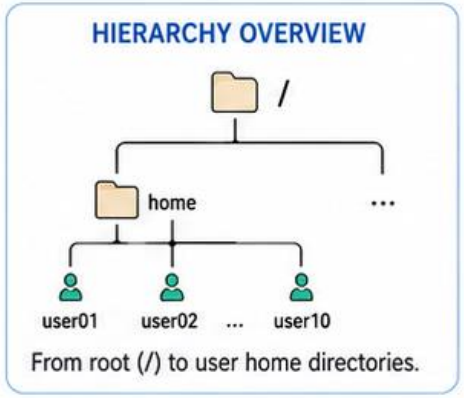
Practice

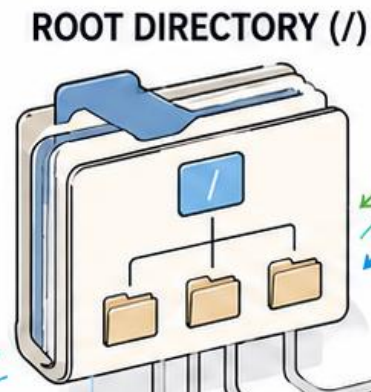
```
ls -l microgen2026/
```

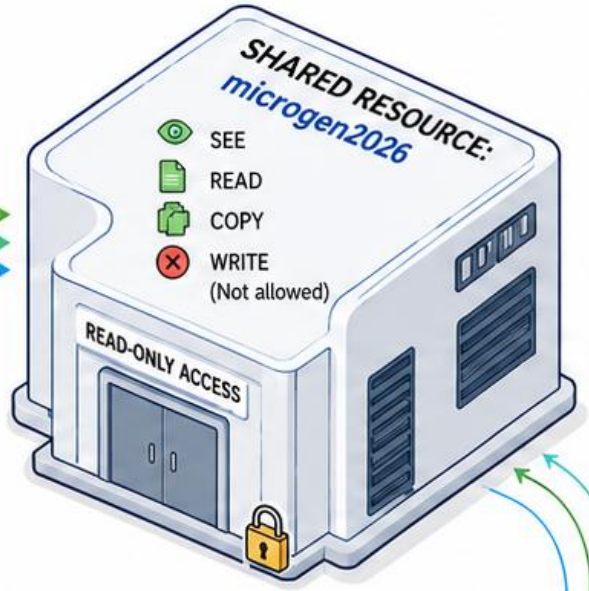
- `ls` is the command (list files).
- `-l` is an option (long format).
- `microgen2026/` is the argument (the folder to look into).

# Linux File System

## Editorial Hierarchy




READ / COPY ACCESS

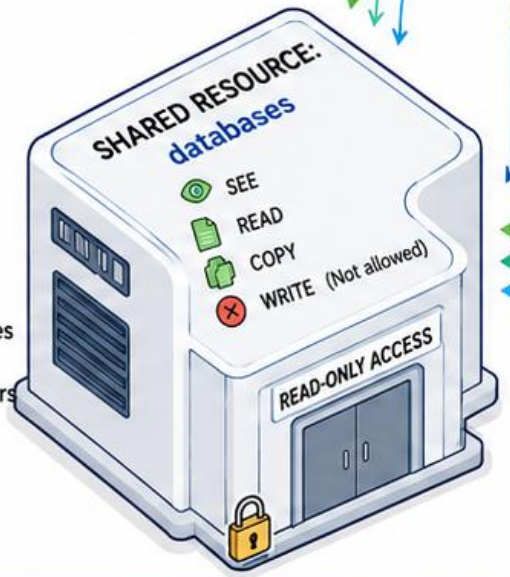
READ-ONLY ACCESS

Shared resources cannot be modified by users (read-only).

ACCESSING SHARED DATA

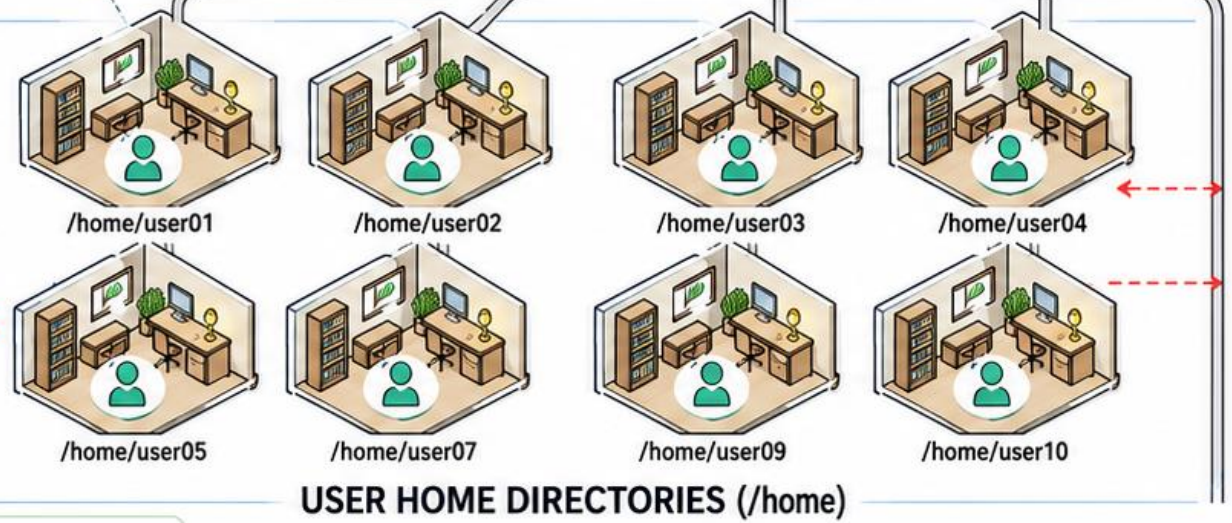
READ / COPY ACCESS

READ-ONLY ACCESS




Shared resources cannot be modified by users (read-only).

ACCESSING SHARED DATA




Stay in their own homes.

Cannot go to someone else's home.

**LEGEND**

- SEE (list / view)
- READ (view contents)
- COPY (copy data)
- WRITE (modify data)
- HIERARCHY / PATH

**READ-ONLY SHARED RESOURCES**

Users can see, read, and copy files from shared resources.
They cannot modify or delete data.

**USER HOME DIRECTORIES**

Each user has a private home directory under /home.
Users manage and modify their own files.

# Let's Practice!

WRAP-UP

# Learning Checklist

- ❑ Connected to your GCP instance on VSCode
- ❑ Know the basic Linux command lines
- ❑ Know how to write and run script from the terminal
- ❑ Downloaded sequencing data to your GCP instance
- ❑ Successful scored ≥ 80% in the Final Quiz

# THE END.

*Ảnh chụp tại đường Trần Phú,*
*TP. HCM,*
*2025.*